

Modelling and System Analysis

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1 Introduction

This report studies the dynamics of a mechanical system composed of two point masses connected by a linear spring, evolving in a plane in the absence of gravity. The mechanical system considered in this work is illustrated in Figure 1.

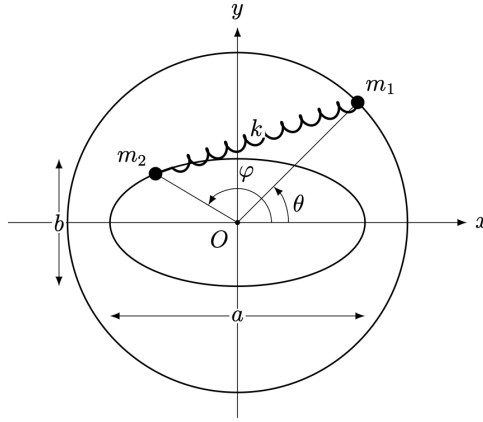


Figure 1: Schematic representation of the mechanical system.

The first mass m_1 is constrained to slide without friction along a circular trajectory of radius R centered at the origin O . The second mass m_2 is constrained to slide on an ellipse, also centered at O , whose major and minor axes are respectively a and b . Unlike m_1 , the mass m_2 is subject to a viscous friction force proportional to its velocity.

The two masses are connected by a linear spring of stiffness k and natural length l_0 , which introduces a coupling between the two trajectories. The system is not subject to gravity, so the only potential energy present in the system is the elastic energy stored in the spring.

The geometric constraints imposed on each mass reduce the system to two degrees of freedom, fully described by the generalized coordinates

$$q_1 = \theta, \quad q_2 = \varphi,$$

where θ denotes the angular position of m_1 on the circle and φ denotes the parametric angle describing the position of m_2 on the ellipse.

This first report successively establishes the kinematic description of the system, derives the Lagrangian, and incorporates the friction force through the generalized force formalism.

2 Kinematic Description

The motion takes place in the plane (x, y) and the origin O is chosen as the reference point for the position vectors.

Mass m_1 . Since m_1 moves on a circle of radius R , its position vector relative to the origin is given by

$$\vec{r}_1 = \begin{pmatrix} R \cos \theta \\ R \sin \theta \end{pmatrix}. \quad (1)$$

Mass m_2 . The mass m_2 moves on an ellipse centered at O with semi-major axis $a/2$ and semi-minor axis $b/2$. A convenient parametric representation of the ellipse leads to the position vector

$$\vec{r}_2 = \begin{pmatrix} \frac{a}{2} \cos \varphi \\ \frac{b}{2} \sin \varphi \end{pmatrix}. \quad (2)$$

Equations (1) and (2) fully describe the spatial configuration of the two masses.

The spring connects the two masses and its deformation depends on the distance between them. The vector connecting m_1 and m_2 is defined as

$$\vec{r}_{12} = \vec{r}_1 - \vec{r}_2. \quad (3)$$

Substituting the expressions of the position vectors given in (1) and (2) into (3) yields

$$\vec{r}_{12} = \begin{pmatrix} R \cos \theta - \frac{a}{2} \cos \varphi \\ R \sin \theta - \frac{b}{2} \sin \varphi \end{pmatrix}. \quad (4)$$

The instantaneous length of the spring corresponds to the Euclidean norm of this vector as

$$l = \|\vec{r}_{12}\| = \sqrt{\left(R \cos \theta - \frac{a}{2} \cos \varphi\right)^2 + \left(R \sin \theta - \frac{b}{2} \sin \varphi\right)^2}. \quad (5)$$

The velocities of the masses are obtained by differentiating the position vectors with respect to time.

Differentiating Equation (1) gives

$$\dot{\vec{r}}_1 = \begin{pmatrix} -R \sin \theta \dot{\theta} \\ R \cos \theta \dot{\theta} \end{pmatrix}. \quad (6)$$

The squared magnitude of this velocity is therefore

$$v_1^2 = R^2 \dot{\theta}^2. \quad (7)$$

Similarly, differentiating Equation (2) leads to

$$\dot{\vec{r}}_2 = \begin{pmatrix} -\frac{a}{2} \sin \varphi \dot{\varphi} \\ \frac{b}{2} \cos \varphi \dot{\varphi} \end{pmatrix}. \quad (8)$$

The squared magnitude of this velocity becomes

$$v_2^2 = \left(\frac{a^2}{4} \sin^2 \varphi + \frac{b^2}{4} \cos^2 \varphi\right) \dot{\varphi}^2. \quad (9)$$

3 Lagrangian Formulation

The kinetic energy of the system is the sum of the kinetic energies of the two masses,

$$K = K_1 + K_2 = \frac{1}{2} m_1 R^2 \dot{\theta}^2 + \frac{1}{2} m_2 \left(\frac{a^2}{4} \sin^2 \varphi + \frac{b^2}{4} \cos^2 \varphi\right) \dot{\varphi}^2. \quad (10)$$

Since the system is not subject to gravity, the only potential energy arises from the deformation of the spring.

The elastic potential energy stored in the spring is

$$P = \frac{1}{2} k (l - l_0)^2. \quad (11)$$

The spring length l appearing in Eq. (11) is given by Eq. (5). This expression introduces a nonlinear coupling between the generalized coordinates θ and φ .

The Lagrangian of the system is defined as the difference between kinetic and potential energies,

$$L = K - P. \quad (12)$$

Substituting the expressions (10) and (11) yields the Lagrangian of the system,

$$L = \frac{1}{2}m_1 R^2 \dot{\theta}^2 + \frac{1}{2}m_2 \left(\frac{a^2}{4} \sin^2 \varphi + \frac{b^2}{4} \cos^2 \varphi \right) \dot{\varphi}^2 - \frac{1}{2}k(l - l_0)^2, \quad (13)$$

where l is the instantaneous spring length defined in (5). The nonlinear coupling between θ and φ appears through l .

The equations of motion will then be obtained by applying the Euler–Lagrange equations,

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = Q_i, \quad i \in \{\theta, \varphi\}. \quad (14)$$

where Q_i represents the generalized forces acting on the system.

In the present system, a viscous friction force acts on the mass m_2 . This friction is assumed to be proportional to the velocity of the mass and opposite to its motion. The friction force is therefore modeled as

$$\vec{F}_f = -\alpha \dot{\vec{r}}_2, \quad (15)$$

where α denotes the viscous damping coefficient and $\dot{\vec{r}}_2$ is the velocity of the mass m_2 , given by equation (8). The generalised force associated with the coordinate φ is obtained via the virtual work principle,

$$Q_\varphi = \vec{F}_f \cdot \frac{\partial \vec{r}_2}{\partial \varphi}. \quad (16)$$

Since $\dot{\vec{r}}_2 = \dot{\varphi} \frac{\partial \vec{r}_2}{\partial \varphi}$, this reduces to

$$Q_\varphi = -\alpha \left(\frac{a^2}{4} \sin^2 \varphi + \frac{b^2}{4} \cos^2 \varphi \right) \dot{\varphi}. \quad (17)$$

This dissipative force introduces an additional term in the Euler–Lagrange equation associated with the coordinate φ . The equations of motion therefore become

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = 0, \quad (18)$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\varphi}} \right) - \frac{\partial L}{\partial \varphi} + \alpha \left(\frac{a^2}{4} \sin^2 \varphi + \frac{b^2}{4} \cos^2 \varphi \right) \dot{\varphi} = 0. \quad (19)$$

Since the friction acts only on the mass m_2 , the coordinate θ is not directly affected by this dissipative force.

4 Equations of Motion

The equations of motion of the system are obtained by applying the Euler–Lagrange equations (14) to the Lagrangian defined in (12).

The Euler–Lagrange equation associated with the coordinate θ gives

$$m_1 R^2 \ddot{\theta} + k(l - l_0) \frac{\partial l}{\partial \theta} = 0. \quad (20)$$

Similarly, applying the Euler–Lagrange equation to the coordinate φ and including the generalised friction force (17) yields

$$\begin{aligned} \frac{d}{dt} \left[m_2 \left(\frac{a^2}{4} \sin^2 \varphi + \frac{b^2}{4} \cos^2 \varphi \right) \dot{\varphi} \right] - m_2 \frac{a^2 - b^2}{4} \sin \varphi \cos \varphi \dot{\varphi}^2 \\ + k(l - l_0) \frac{\partial l}{\partial \varphi} + \alpha \left(\frac{a^2}{4} \sin^2 \varphi + \frac{b^2}{4} \cos^2 \varphi \right) \dot{\varphi} = 0. \end{aligned} \quad (21)$$

Equations (20) and (21) form a system of two nonlinear coupled differential equations describing the dynamics of the system.

4.1 Derivatives of the spring length

To make the equations of motion explicit, the derivatives of the spring length l with respect to the generalized coordinates must be computed.

From (5), the derivative with respect to θ becomes

$$\frac{\partial l}{\partial \theta} = \frac{-R \sin \theta \left(R \cos \theta - \frac{a}{2} \cos \varphi \right) + R \cos \theta \left(R \sin \theta - \frac{b}{2} \sin \varphi \right)}{l}. \quad (22)$$

Similarly, the derivative with respect to φ is

$$\frac{\partial l}{\partial \varphi} = \frac{\frac{a}{2} \sin \varphi \left(R \cos \theta - \frac{a}{2} \cos \varphi \right) - \frac{b}{2} \cos \varphi \left(R \sin \theta - \frac{b}{2} \sin \varphi \right)}{l}. \quad (23)$$

4.2 Final Equations of Motion

Substituting the derivatives into the Euler-Lagrange equations yields the explicit equations of motion.

$$m_1 R^2 \ddot{\theta} + \frac{k(l-l_0)}{l} \left[-R \sin \theta \left(R \cos \theta - \frac{a}{2} \cos \varphi \right) + R \cos \theta \left(R \sin \theta - \frac{b}{2} \sin \varphi \right) \right] = 0 \quad (24)$$

$$\begin{aligned} m_2 \left(\frac{a^2}{4} \sin^2 \varphi + \frac{b^2}{4} \cos^2 \varphi \right) \ddot{\varphi} + m_2 \frac{a^2 - b^2}{4} \sin \varphi \cos \varphi \dot{\varphi}^2 \\ + \frac{k(l-l_0)}{l} \left[\frac{a}{2} \sin \varphi \left(R \cos \theta - \frac{a}{2} \cos \varphi \right) - \frac{b}{2} \cos \varphi \left(R \sin \theta - \frac{b}{2} \sin \varphi \right) \right] + \alpha \left(\frac{a^2}{4} \sin^2 \varphi + \frac{b^2}{4} \cos^2 \varphi \right) \dot{\varphi} = 0 \end{aligned} \quad (25)$$

Equations (24) and (25) constitute the complete nonlinear equations of motion of the system. These equations describe the coupled dynamics of the two masses connected by the spring, including the viscous friction acting on the second mass.

5 Non-dimensionalization

Before proceeding to non-dimensionalization, the geometric parameters by the problem statement,

$$a = \frac{3}{4} R, \quad b = \frac{1}{4} R, \quad l_0 = R,$$

are substituted into the final equations of motion derived in the previous section. The position vectors become

$$\vec{r}_1 = \begin{pmatrix} R \cos \theta \\ R \sin \theta \end{pmatrix}, \quad (26)$$

$$\vec{r}_2 = \begin{pmatrix} \frac{3R}{8} \cos \varphi \\ \frac{R}{8} \sin \varphi \end{pmatrix}. \quad (27)$$

The instantaneous spring length is given by

$$\begin{aligned} l &= \sqrt{\left(R \cos \theta - \frac{3R}{8} \cos \varphi \right)^2 + \left(R \sin \theta - \frac{R}{8} \sin \varphi \right)^2} \\ &= R \sqrt{1 - \frac{3}{4} \cos \theta \cos \varphi - \frac{1}{4} \sin \theta \sin \varphi + \frac{1}{64} (9 \cos^2 \varphi + \sin^2 \varphi)}. \end{aligned} \quad (28)$$

Substituting the geometric parameters into equation (24) and simplifying, the equation of motion for θ reduces to

$$\boxed{m_1 R^2 \ddot{\theta} + \frac{k R^2}{8} \frac{l - R}{l} (3 \sin \theta \cos \varphi - \cos \theta \sin \varphi) = 0.} \quad (29)$$

Similarly, substituting the geometric parameters into equation (25) and simplifying, the equation of motion for φ reduces to

$$\boxed{\begin{aligned} & \frac{m_2 R^2}{64} (9 \sin^2 \varphi + \cos^2 \varphi) \ddot{\varphi} + \frac{m_2 R^2}{8} \sin \varphi \cos \varphi \dot{\varphi}^2 \\ & + \frac{k R^2}{8} \frac{l-R}{l} (3 \sin \varphi \cos \theta - \cos \varphi \sin \theta - \sin \varphi \cos \varphi) \\ & + \frac{\alpha R^2}{64} (9 \sin^2 \varphi + \cos^2 \varphi) \dot{\varphi} = 0. \end{aligned}} \quad (30)$$

In order to reduce the number of independent parameters and reveal the characteristic scales governing the dynamics, the equations of motion are now written in dimensionless form. The natural length scale of the system is the radius R of the circular track, which provides the characteristic geometric length of the problem. The dimensionless time variable is introduced as

$$\tau = \omega_0 t, \quad \text{with} \quad \omega_0 = \sqrt{\frac{k}{m_1}}. \quad (31)$$

The quantity ω_0 corresponds to the natural frequency of a mass-spring oscillator with mass m_1 and stiffness k , which sets the characteristic time scale of the system. With this choice, the time derivatives transform according to

$$(\dot{\cdot}) = \omega_0 (\cdot)', \quad (\ddot{\cdot}) = \omega_0^2 (\cdot)'',$$

where the prime notation $(\cdot)'$ denotes differentiation with respect to the dimensionless time τ .

Two dimensionless parameters naturally emerge from this process. The first is the mass ratio

$$\mu = \frac{m_2}{m_1}, \quad (32)$$

which characterizes the relative inertia of the two particles. When $\mu \ll 1$, the second mass responds almost instantaneously to the spring force and the dynamics are dominated by the motion of m_1 . Conversely, when $\mu \gg 1$, the elliptical mass is much heavier and the circular mass acts as a light oscillator driven by the motion of m_2 .

The second parameter is the dimensionless friction coefficient

$$\beta = \frac{\alpha}{m_1 \omega_0} = \frac{\alpha}{\sqrt{k m_1}}, \quad (33)$$

which measures the strength of the viscous dissipation acting on m_2 relative to the elastic restoring force. The limit $\beta = 0$ corresponds to the frictionless system, while large values of β indicate that the motion of m_2 is strongly damped.

With this choice of reference quantities, the common factor $m_1 R^2 \omega_0^2 = k R^2$ can be factored out of every term in equations (29) and (30), leaving the system expressed solely in terms of μ , β and the dimensionless time τ .

The dimensionless equation of motion for θ reads

$$\boxed{\theta'' + \frac{1}{8} \frac{l-R}{l} (3 \sin \theta \cos \varphi - \cos \theta \sin \varphi) = 0.} \quad (34)$$

Proceeding similarly with equation (30), the dimensionless equation of motion for φ reads

$$\boxed{\begin{aligned} & \frac{\mu}{64} (9 \sin^2 \varphi + \cos^2 \varphi) \varphi'' + \frac{\mu}{8} \sin \varphi \cos \varphi (\varphi')^2 \\ & + \frac{1}{8} \frac{l-R}{l} (3 \sin \varphi \cos \theta - \cos \varphi \sin \theta - \sin \varphi \cos \varphi) \\ & + \frac{\beta}{64} (9 \sin^2 \varphi + \cos^2 \varphi) \varphi' = 0. \end{aligned}} \quad (35)$$

In both equations, the ratio $(l-R)/l$ involves the dimensionless spring length l/R , which is directly obtained from equation (28) as

$$\frac{l}{R} = \sqrt{1 - \frac{3}{4} \cos \theta \cos \varphi - \frac{1}{4} \sin \theta \sin \varphi + \frac{1}{64} (9 \cos^2 \varphi + \sin^2 \varphi)}, \quad (36)$$

so that the dimensionless spring extension can be evaluated as

$$\frac{l-R}{l} = 1 - \frac{1}{l/R}. \quad (37)$$

The non-dimensionalization reveals that the dynamics of the system are entirely governed by only two parameters: the mass ratio μ and the damping coefficient β .

Finally, the dimensionless total energy of the system is obtained from equations (10) and (11) following the same procedure. This gives

$$\hat{E} = \frac{1}{2}(\theta')^2 + \frac{\mu}{128} (9 \sin^2 \varphi + \cos^2 \varphi) (\varphi')^2 + \frac{1}{2} \left(\frac{l}{R} - 1 \right)^2. \quad (38)$$

Each term is non-negative, so that $\hat{E} \geq 0$ at all times. In the absence of friction ($\beta = 0$), the total energy $\hat{E}$ is conserved.

When friction is present ($\beta > 0$), the rate of change of the total energy can be derived by multiplying the dimensionless equation of motion for θ (34) by θ' and the dimensionless equation of motion for φ (35) by φ' . Starting with the equation for θ ,

$$\theta' \theta'' + \frac{1}{8} \frac{l-R}{l} (3 \sin \theta \cos \varphi - \cos \theta \sin \varphi) \theta' = 0. \quad (39)$$

The first term can be recognized as

$$\theta' \theta'' = \frac{d}{d\tau} \left(\frac{1}{2} (\theta')^2 \right), \quad (40)$$

which is the time derivative of the dimensionless kinetic energy of m_1 . Similarly, multiplying the equation for φ by φ' gives

$$\begin{aligned} & \frac{\mu}{64} (9 \sin^2 \varphi + \cos^2 \varphi) \varphi' \varphi'' + \frac{\mu}{8} \sin \varphi \cos \varphi (\varphi')^3 \\ & + \frac{1}{8} \frac{l-R}{l} (3 \sin \varphi \cos \theta - \cos \varphi \sin \theta - \sin \varphi \cos \varphi) \varphi' \\ & + \frac{\beta}{64} (9 \sin^2 \varphi + \cos^2 \varphi) (\varphi')^2 = 0. \end{aligned} \quad (41)$$

The first two terms correspond to the time derivative of the dimensionless kinetic energy of m_2 ,

$$\frac{d}{d\tau} \left[\frac{\mu}{128} (9 \sin^2 \varphi + \cos^2 \varphi) (\varphi')^2 \right] = \frac{\mu}{64} (9 \sin^2 \varphi + \cos^2 \varphi) \varphi' \varphi'' + \frac{\mu}{8} \sin \varphi \cos \varphi (\varphi')^3, \quad (42)$$

where the second contribution arises from differentiating the φ -dependent geometric factor. Furthermore, the spring-related terms in equations (39) and (41) combine to give the time derivative of the dimensionless potential energy,

$$\frac{d}{d\tau} \left[\frac{1}{2} \left(\frac{l}{R} - 1 \right)^2 \right] = \frac{l-R}{l} \left[\frac{1}{8} (3 \sin \theta \cos \varphi - \cos \theta \sin \varphi) \theta' + \frac{1}{8} (3 \sin \varphi \cos \theta - \cos \varphi \sin \theta - \sin \varphi \cos \varphi) \varphi' \right]. \quad (43)$$

Adding equations (39) and (41) and using the identities (40), (42) and (43), all conservative terms regroup into $d\hat{E}/d\tau$ and only the friction term remains. This yields

$$\boxed{\frac{d\hat{E}}{d\tau} = -\frac{\beta}{64} (9 \sin^2 \varphi + \cos^2 \varphi) (\varphi')^2 \leq 0}, \quad (44)$$

which confirms that the viscous force acting on m_2 is the only source of dissipation in the system. The inequality follows from the fact that the geometric factor $(9 \sin^2 \varphi + \cos^2 \varphi)$ is strictly positive for all values of φ .

6 Numerical Resolution and Analysis

The dimensionless equations of motion (34) and (35) do not admit closed-form solutions. This section integrates them numerically, validates the implementation, and investigates the system dynamics. All simulations use $\mu = 1$ (equal masses) unless stated otherwise.

6.1 State-space formulation and integration method

The system is recast as a first-order system $\mathbf{y}' = \mathbf{f}(\mathbf{y})$ by introducing the state vector

$$\mathbf{y} = \begin{pmatrix} \theta \\ \varphi \\ \theta' \\ \varphi' \end{pmatrix}, \quad (45)$$

where the prime denotes differentiation with respect to τ . The right-hand side of this system is

$$\mathbf{f}(\mathbf{y}) = \begin{pmatrix} \theta' \\ \varphi' \\ F_\theta(\theta, \varphi) \\ F_\varphi(\theta, \varphi, \varphi') \end{pmatrix}, \quad (46)$$

where the first two components are the velocities already present in $\mathbf{y}$, and the last two are the accelerations F_θ and F_φ , obtained by isolating θ'' and φ'' from the dimensionless equations of motion (34) and (35). Defining the geometric factor $g(\varphi) = 9 \sin^2 \varphi + \cos^2 \varphi > 0$, these read

$$F_\theta = -\frac{1}{8} \frac{l-R}{l} (3 \sin \theta \cos \varphi - \cos \theta \sin \varphi), \quad (47)$$

$$F_\varphi = -\frac{64}{\mu g(\varphi)} \left[\frac{\mu}{8} \sin \varphi \cos \varphi (\varphi')^2 + \frac{1}{8} \frac{l-R}{l} (3 \sin \varphi \cos \theta - \cos \varphi \sin \theta - \sin \varphi \cos \varphi) + \frac{\beta}{64} g(\varphi) \varphi' \right]. \quad (48)$$

The system is integrated with the adaptive RK45 method from `scipy.integrate.solve_ivp` [1], with `rtol` = 10^{-10} and `atol` = 10^{-12} . An interactive web-based simulator¹ was developed in parallel and used to explore the parameter space and select the initial conditions.

6.2 Initial conditions

All simulations use the same initial conditions:

$$\theta_0 = \frac{\pi}{2} \text{ rad}, \quad \varphi_0 = 0 \text{ rad}, \quad \theta'_0 = 0.5 \text{ rad}/\tau, \quad \varphi'_0 = 4.0 \text{ rad}/\tau, \quad (49)$$

where positive angular velocities correspond to counter-clockwise motion in the (x, y) plane. The initial energy is $\hat{E}_0 \approx 0.252$, split equally between $K_1 \approx 50\%$ and $K_2 \approx 50\%$, with a small contribution from the spring ($P \approx 1\%$). This balanced partition ensures that neither mass dominates the dynamics and that both the direct dissipation channel (through K_2) and the indirect channel (through the spring coupling, $K_1 \rightarrow P \rightarrow K_2$) are active from the start.

6.3 Validation: energy conservation

Figure 2 monitors the total energy and its relative drift for $\beta = 0$.

Observation. The energy remains constant at $\hat{E}_0 = 0.2523$ over the entire simulation. The relative drift $(\hat{E} - \hat{E}_0)/\hat{E}_0$ stays below 5×10^{-8} .

Conclusion. Since the analytical derivation in Section 5 guarantees $d\hat{E}/d\tau = 0$ when $\beta = 0$, the near-perfect conservation of $\hat{E}$ in the numerical solution confirms that the equations of motion have been correctly implemented and that the state-space formulation (46)–(48) is consistent with the Lagrangian model.

¹https://valentinangenot.github.io/simulation_modelling/

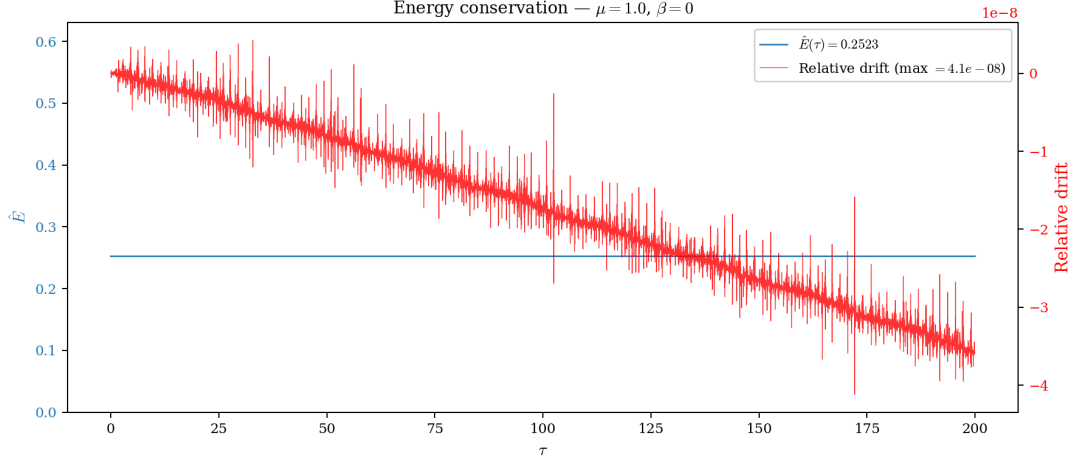


Figure 2: Energy conservation ($\mu = 1$, $\beta = 0$). Blue: total energy (constant). Red: relative drift ($< 5 \times 10^{-8}$).

6.4 System response and energy exchanges

Figure 3 compares the conservative ($\beta = 0$) and dissipative ($\beta = 0.5$) responses.

Observation — conservative case (left). Both angles grow monotonically: the masses rotate continuously on their constraint curves. Over $\tau = 200$, m_1 completes approximately 16 full turns ($\theta \approx 100$ rad) while m_2 completes about 56 turns ($\varphi \approx 350$ rad). The faster rotation of m_2 reflects both the higher initial velocity ($\varphi'_0 = 4$ vs. $\theta'_0 = 0.5$) and the smaller size of the ellipse, which requires less displacement per radian. Despite the continuous rotation, the total energy $\hat{E}$ remains constant, while K_1 , K_2 and P oscillate: each time the angular separation between the two masses changes, the spring stretches or compresses, periodically converting kinetic energy into potential energy and vice versa.

Observation — dissipative case (right). With $\beta = 0.5$, both angles initially grow but their behaviour differs markedly from the conservative case. The mass m_2 does not rotate continuously: φ oscillates back and forth with decreasing amplitude, reversing direction several times before converging to a final position ($\varphi \rightarrow 6$ rad). The mass m_1 undergoes a few turns ($\theta \rightarrow 22$ rad) before also coming to rest. The total energy decays to zero within approximately $\tau \approx 100$. The energy panel reveals that K_2 is dissipated first, since the friction acts directly on m_2 . The kinetic energy K_1 of the frictionless mass persists longer, decaying only as the spring coupling progressively transfers it to m_2 . The potential energy P vanishes last, once both masses have stopped and the spring has relaxed.

Conclusion. In the conservative case, the spring does not prevent the masses from rotating, but periodically modulates their velocities, producing the oscillations visible in the energy decomposition. In the dissipative case, friction on m_2 first arrests its rotation, then the spring coupling gradually extracts the remaining kinetic energy from m_1 until both masses come to rest. The indirect dissipation pathway $K_1 \rightarrow P \rightarrow K_2 \rightarrow$ friction is visible in the energy panel: K_1 persists longer than K_2 because it can only be dissipated through the spring coupling.

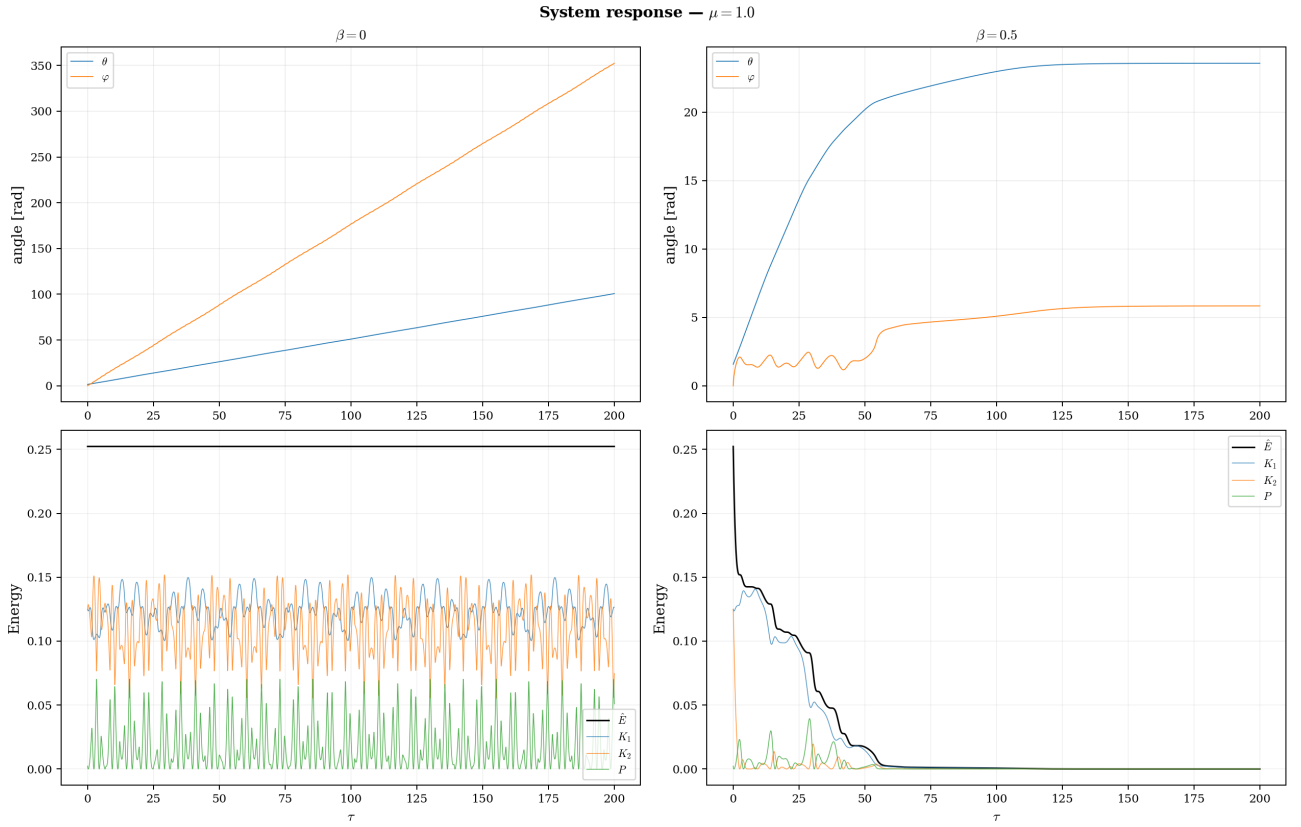


Figure 3: System response ($\mu = 1$). Left: $\beta = 0$. Right: $\beta = 0.5$. Top: angular positions. Bottom: energy decomposition.

6.5 Influence of the damping coefficient β

Figure 4 shows the total energy evolution for six values of β ranging from 0 to 20.

Observation. For $\beta = 0$, the energy is constant. For $\beta = 0.1$ and 0.5 , the energy decays efficiently toward zero. For $\beta = 2$, the decay is slower. For $\beta = 8$ and 20 , a significant fraction of the initial energy persists at $\tau = 200$.

Interpretation. This counter-intuitive result, where more friction leads to *less* dissipation, arises because the friction acts only on m_2 . For the 50% of energy initially stored in K_1 to be dissipated, it must first be transferred to m_2 through the spring coupling. When β is moderate, m_2 is free to oscillate in response to the spring force, and each oscillation cycle dissipates a fraction of the energy received. When β is large, the friction force on m_2 dominates the spring force: m_2 is rapidly immobilized and remains nearly at rest ($\varphi' \approx 0$). The spring still exerts a force on m_2 , but the friction prevents it from accelerating. Since the dissipation power is proportional to $g(\varphi) (\varphi')^2$, a nearly stationary m_2 dissipates almost no energy, and the kinetic energy of m_1 remains trapped.

Conclusion. There exists an optimal range of β that allows efficient dissipation of the total energy. This is a structural property of systems where damping is localised on a subset of the degrees of freedom.

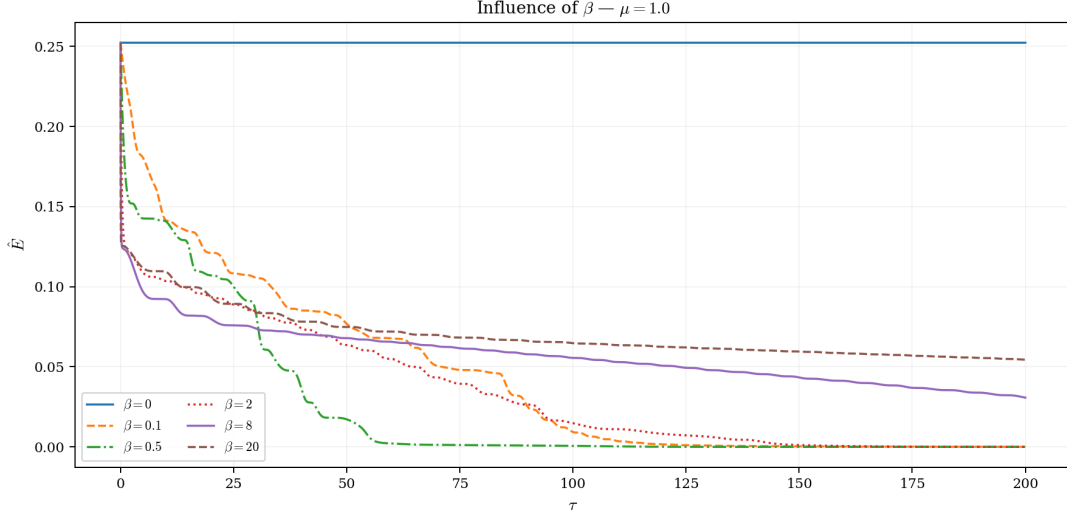


Figure 4: Influence of β on total energy ($\mu = 1$). Moderate β dissipates efficiently; large β traps energy by immobilizing m_2 .

6.6 Influence of the mass ratio μ

Figure 5 shows the energy evolution for four values of μ at $\beta = 0.5$.

Observation. For $\mu = 0.1, 0.5, 1$ and 2 , the energy decays to zero within approximately $\tau \approx 50$. All three curves exhibit the same qualitative behaviour; only the initial energy level differs. Starting at $\mu = 5$, the dissipation becomes visibly slower and does not reach zero within the simulation window.

Interpretation. The mass ratio μ affects the dissipation through two competing mechanisms. First, the initial energy increases with μ because $K_2 = \frac{\mu}{128} g(\varphi_0) (\varphi'_0)^2$ is proportional to μ , so there is more energy to dissipate. Second, the effective inertia of m_2 on the ellipse scales as $\frac{\mu}{64} g(\varphi)$, which is the coefficient of φ'' in equation (35). A larger μ increases this inertia, reducing the angular acceleration that the spring force can impart to m_2 . The result is a lower φ' for a given spring force, and therefore a lower instantaneous dissipation power $\frac{\beta}{64} g(\varphi) (\varphi')^2$. Conversely, a light m_2 ($\mu \ll 1$) accelerates easily, reaches high angular velocities, and dissipates energy rapidly. For $\mu = 2, 5$ and 10 , full dissipation is guaranteed by the analytical result (44) ($d\dot{E}/d\tau \leq 0$ for any $\beta > 0$), but requires a simulation time well beyond $\tau = 200$.

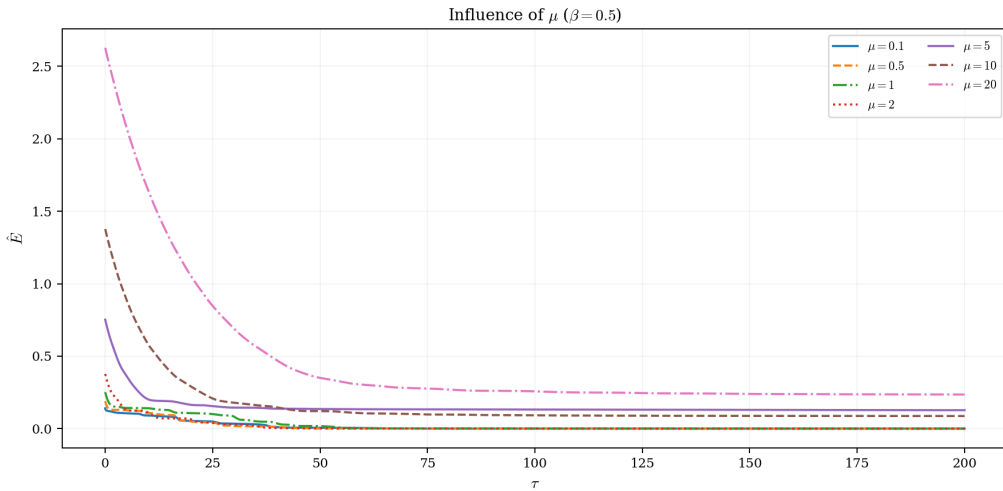


Figure 5: Influence of μ on total energy ($\beta = 0.5$). Lighter m_2 dissipates energy faster.

6.7 Phase portraits

Figure 6 compares the phase-space trajectories for $\beta = 0$ and $\beta = 0.5$.

Observation — conservative case (left). Since the masses rotate continuously, the phase portraits are not closed orbits but horizontal bands swept from left to right. For m_1 (top left), the angular velocity θ' oscillates within a narrow band ($\approx 0.45\text{--}0.55$), modulated by the spring coupling. For m_2 (bottom left), the band is much wider ($\varphi' \approx 1.2\text{--}4.2$), reflecting stronger velocity fluctuations due to the smaller inertia of m_2 on the ellipse.

Conclusion. In the dissipative case, friction decelerates m_2 first: it reverses direction several times, oscillating back and forth on the ellipse until it stops. The mass m_1 , with no friction, decelerates more slowly through the spring coupling and covers progressively shorter arcs before also coming to rest. The final rest positions depend on the initial conditions.

Conclusion. The phase portraits reveal the asymmetry of the dissipation mechanism. The mass m_2 , subject to friction, loses most of its velocity quickly (the steep initial drop in φ'). The mass m_1 , with no friction, decelerates gradually as the spring coupling slowly drains its kinetic energy. The small loops visible for m_2 near the origin correspond to the back-and-forth oscillations observed in the time-domain response (Figure 3).

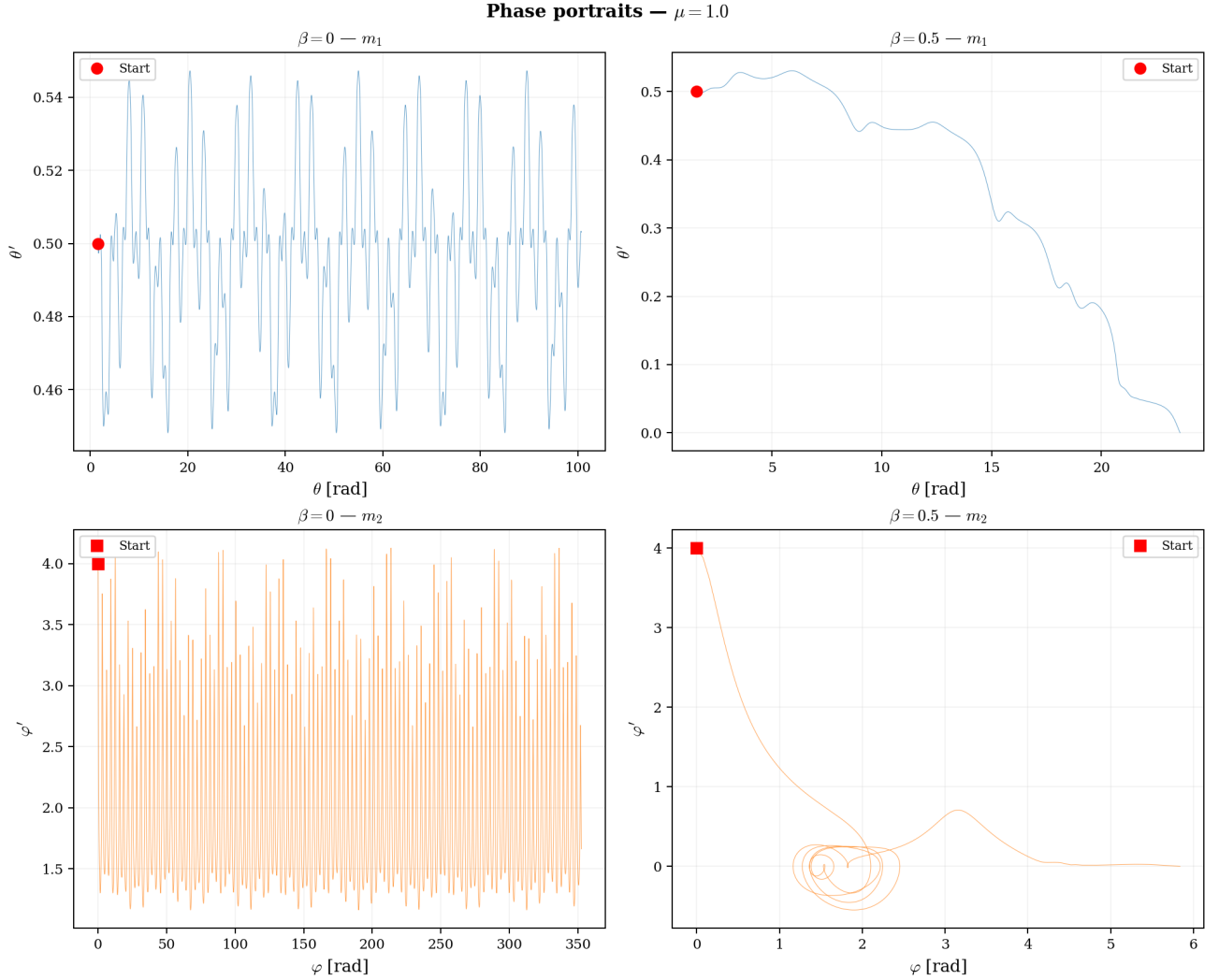


Figure 6: Phase portraits ($\mu = 1$). Left: $\beta = 0$ (dense quasi-periodic orbits). Right: $\beta = 0.5$ (inward spirals). Top: m_1 . Bottom: m_2 .

6.8 Spatial trajectories

Figure 7 shows the physical paths of both masses in the (x, y) plane. The line thickness is proportional to the time spent at each location: thicker segments indicate regions where the mass passes frequently or slows down.

Observation — conservative case (a). Both masses cover their entire constraint curves with roughly uniform thickness, indicating a nearly constant angular velocity throughout the simulation.

Observation — dissipative case (b). The thickness is no longer uniform. For m_2 , the ellipse shows a thick accumulation near the final rest position, where the mass oscillates back and forth with decreasing amplitude

before stopping. For m_1 , the circle shows a similar thickening on the arc where the mass decelerates and eventually comes to rest.

Conclusion. The density map confirms the dissipation sequence observed in the previous figures: m_2 is arrested first by friction and accumulates at a fixed point on the ellipse, while m_1 decelerates more slowly through the spring coupling and concentrates on a narrowing arc of the circle.

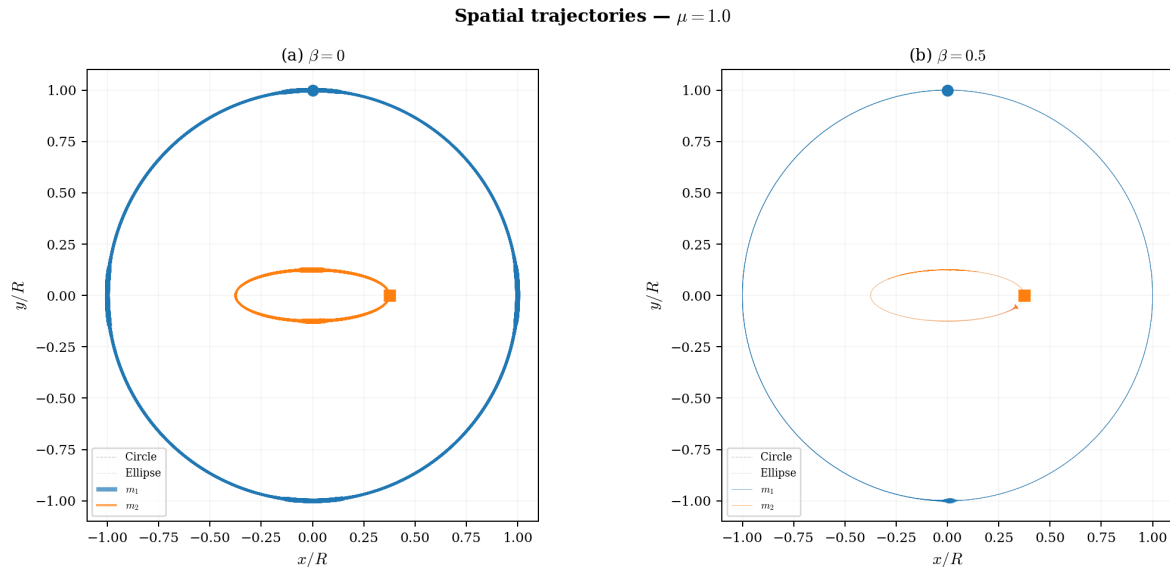


Figure 7: Spatial trajectories ($\mu = 1$). (a) $\beta = 0$: both masses explore their constraint curves. (b) $\beta = 0.5$: convergence to rest.

References

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